

Sales Summary from a Tiny SQLite Database using Python

Source code:

```
sqlite.py - C:/Users/Sathvikaa Sri Kumar/AppData/Local/Programs/Python/Python311/sqlite.py (3.11.2)
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import sqlite3
import matplotlib.pyplot as plt

# Step 1: Connect to or create the SQLite database
conn = sqlite3.connect('sales_data.db')
cursor = conn.cursor()

# Step 2: Create a sales table
cursor.execute('''
    CREATE TABLE IF NOT EXISTS sales (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        date TEXT,
        product TEXT,
        quantity INTEGER,
        price REAL
    )
''')

# Step 3: Insert updated sample data (if table is empty)
cursor.execute('SELECT COUNT(*) FROM sales')
if cursor.fetchone()[0] == 0:
    sample_data = [
        ('2025-05-05', 'Headphones', 4, 150.00),
        ('2025-05-05', 'Monitor', 2, 300.00),
        ('2025-05-06', 'Keyboard', 6, 90.00),
        ('2025-05-06', 'Mouse', 8, 40.00),
        ('2025-05-07', 'Monitor', 1, 300.00),
        ('2025-05-07', 'Headphones', 3, 150.00),
    ]
    cursor.executemany('INSERT INTO sales (date, product, quantity, price) VALUES (?, ?, ?, ?)', sample_data)
    conn.commit()

# Step 4: Run SQL query - total revenue by product
cursor.execute('''
    SELECT product, SUM(quantity * price) AS revenue
    FROM sales
    GROUP BY product
''')
results = cursor.fetchall()
print("Total Revenue by Product:")
for row in results:
    print(f"{row[0]}: ${row[1]:.2f}")

# Step 5: Plot bar chart using matplotlib
products = [row[0] for row in results]
revenues = [row[1] for row in results]
```

```

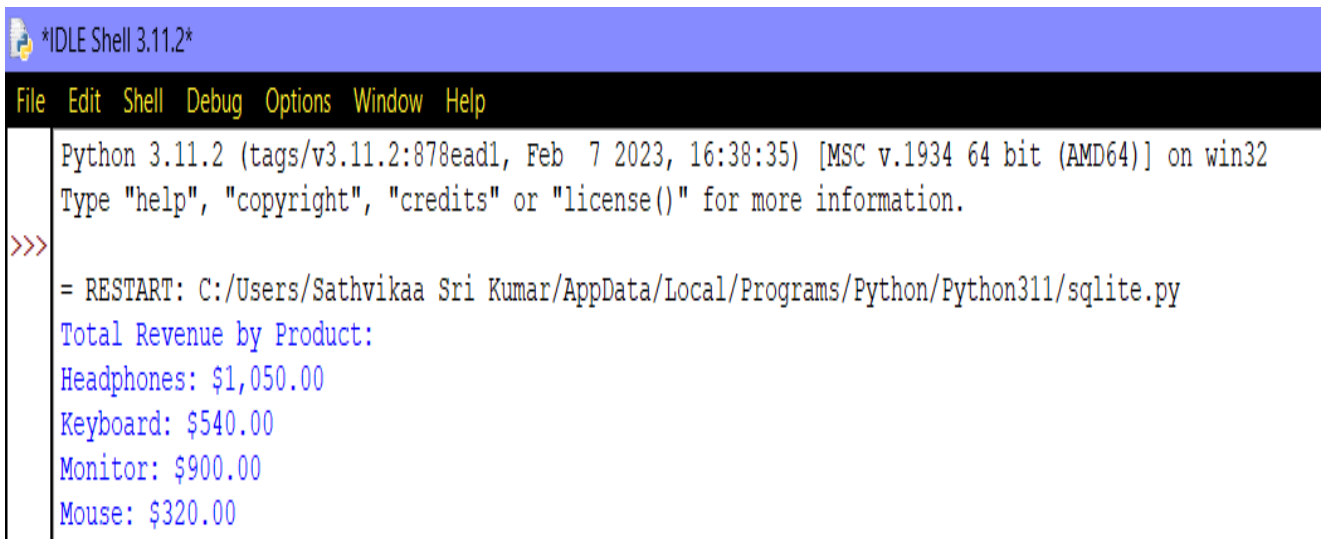
# Step 5: Plot bar chart using matplotlib
products = [row[0] for row in results]
revenues = [row[1] for row in results]

plt.figure(figsize=(6,4))
plt.bar(products, revenues, color='skyblue')
plt.title('Total Revenue by Product')
plt.xlabel('Product')
plt.ylabel('Revenue ($)')
plt.tight_layout()
plt.show()

# Close the connection
conn.close()

```

Output:



```

Python 3.11.2 (tags/v3.11.2:878ead1, Feb 7 2023, 16:38:35) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sathvikaa Sri Kumar/AppData/Local/Programs/Python/Python311/sqlite.py
Total Revenue by Product:
Headphones: $1,050.00
Keyboard: $540.00
Monitor: $900.00
Mouse: $320.00

```

Total Revenue by Product

